

Predicting Fantasy Football Points on Prior Year Player Statistics

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1 Introduction

Participating in Fantasy Football is a yearly tradition for many fans of the NFL. At the beginning of the football season, fans "draft" their fantasy teams from the list of active players on NFL teams. They then watch as those players either bring them glory or defeat for their weekly matchup against another member of the league. This weekly play, culminates in a knockout bracket to crown the winner on the season [1]. This begs the question, how can we draft the best team possible so that we can win?

This question applies not only to fantasy football drafts, but real life sports drafting and scouting. Through use of data collection, analysis, interviews, and gut instinct, many professional sports teams conduct analyses of potential prospects to then draft onto their own teams [2]. In fantasy football's case, we can also utilize data collection and analyses to predict which players will provide the most return on the season based on last years statistics.

2 Data Source & Tidying

The data that was used in conducting these analyses was sourced from the *Pro Football Reference* website. This site contains information about every football player that had played in the NFL since 1970; their real life statistics like passing yards, rushing yards, or receiving yards, as well as their total fantasy points for the season. For this analyses, we use the data from the past 20 years; 2004-2023 for training, and 2024 to predict, and have 12724 players. This information is a great starting point in providing data points for finding trends based on player performance. The data utilized from the site was scraped from each year's table and combined into one complete dataset. However, this information needed to be cleaned up in order to be usable.

Each year's table contained a header row to distinguish which column belonged to which player role. I needed to combine them to have usable single-column headers. For instance, there were "Yds" under "Passing", "Rushing", as well as "Receiving", so I needed to combine each to make a distinct column of "PYds", "RushYds", and "RecYds" respectively. There were also columns that were not useful to my analyses that I could remove. I also added a 'years_played' column, that counted the number of times a player appeared in the data starting from their first year. In order to predict next year's fantasy points based on the stats of the previous year, we must also add the column "NxtYrFantPt", which can be used when looking at our predictions for each player.

Once the columns were dealt with, the rest of the data could be adjusted. In cleaning the numeric data, there were many stats that were NA, which I replaced with zero values. I also updated the type of data in each column to reflect whether it was a string or a number. We also need to filter out any player that has only played for one year; as they would not have any fantasy statistics for the next year to compare to. This included rookies, retirees, or any player that had left the NFL. Once all the data were tidy, they could be used in modeling.

Example Stats																			
Year	Player	Tm	FantPos	Age	G	RAtt	RushYds	Y/A	RushTD	Tgt	Rec	RecYds	Y/R	RecTD	Fmb	FL	2PM	2PP	FantPt
2024	Lamar Jackson	BAL	QB	27	17	139	915	6.58	4	0	0	0	0	0	10	5	1	0	430
2006	LaDainian Tomlinson	SDG	RB	27	16	348	1815	5.22	28	80	56	508	9.07	3	2	1	0	0	425
2018	Patrick Mahomes	KAN	QB	23	16	60	272	4.53	2	0	0	0	0	0	9	2	1	0	417
2022	Patrick Mahomes	KAN	QB	27	17	61	358	5.87	4	1	1	6	6	0	5	0	1	2	416
2019	Lamar Jackson	BAL	QB	22	15	176	1206	6.85	7	0	0	0	0	0	9	2	0	0	416

Figure 1: Example of Highest Fantasy Point Players

3 Methods

In trying to predict which players would be the best to draft for fantasy, we want to look at fantasy points for the season. Players who provide the most fantasy points throughout the season will give you a greater chance of winning week-to-week matches and ultimately the entire tournament. However, each position in fantasy football is measured differently. A quarterback gains fantasy points for throwing and completing longer passes, whereas a running back may gain more for longer runs they complete. Thus, each player and their corresponding stats would need to be separated to analyze their positions.

Since fantasy points is a continuous value we are trying to predict, we will be looking at different regression models and seeing which predict the next year’s fantasy points most accurately. For this comparison, we will be looking at linear regression; random forest with bagging, boosting, k nearest neighbors, and a neural net model. In all but the linear model and neural net model, we also use cross-validation to identify the best model parameters that can be used that minimize our error. We will also be standardizing the data for linear, KNN, and neural net. This ensures that the sensitivity to the scale of the covariate values will not impact the predictions as much. For the tree methods, this is not as important as they break up based on thresholds.

In determining which models perform the best we will look at the R squared, root mean squared error, and mean squared error. The R squared value allows us to tell whether or not the model fits the data well. Looking at the RMSE and MSE lets us see how far off our predictions are from the true values.

4 Results

For the purposes of brevity, we will be looking mostly at running back (RB) models and their performance. In Fig 1, we can see the MSE, R2, and RMSE of each of our models used to predict running back performance for the next year. For models not including KNN, starting with the R2 value, we can see that none of the models approach 1, indicating none of them are a great fit for the data. All but K-NN are close, between 0.42 and 0.47, also telling us that none of them are significantly better than the other at predicting with the data provided. If we look at the RMSE, the mean of the error is around 60, which is a high value, considering the highest amount of points for a RB in 2024 was 322, and on the lower end it would be a wide range of values. KNN might be performing poorly due to a larger number of covariates, as well as some that may just be adding noise.

RB Model Eval Metrics					
	Linear Regression	Random Forest	Boosting	K Nearest Neighbors	Neural Net
MSE	3620.843857	3795.629982	3862.068155	4053.994018	3657.582503
R2	0.463875	0.437995	0.428158	0.399740	0.458436
RMSE	60.173448	61.608684	62.145540	63.670983	60.477951

Figure 2: RB Model Evaluation Metrics

This was slightly puzzling, as with prior knowledge of football I would have assumed that some of the statistics provided would give a better understanding of where they would end in total fantasy points. Looking into which covariates were most important for each model, we can identify that "Rushing Yards" is the highest valued one for all. I also wanted to investigate "Age" as it is a highly discussed variable in football performance. If a player is older, their athletic abilities can fall off or make them prone to an injury. In Fig 2 and 3, we can observe the trends of these statistics in predicting the next year of fantasy points. Figure 2 shows decent upwards sloping trend in predicting the next year's fantasy points based on rushing yards. Figure 3 shows that age also has a slight downwards sloping trend, indicating there is some relationship between age and next year's fantasy points.

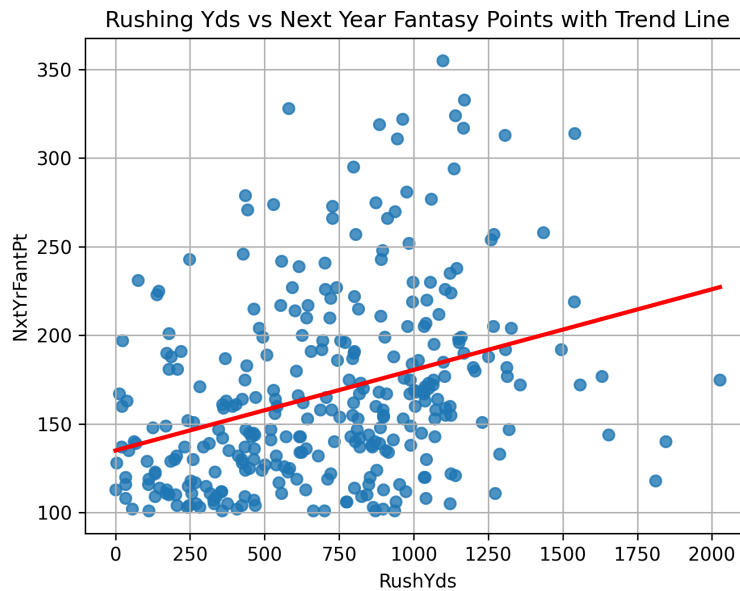


Figure 3: Rush Yards vs Next Year Fantasy Points

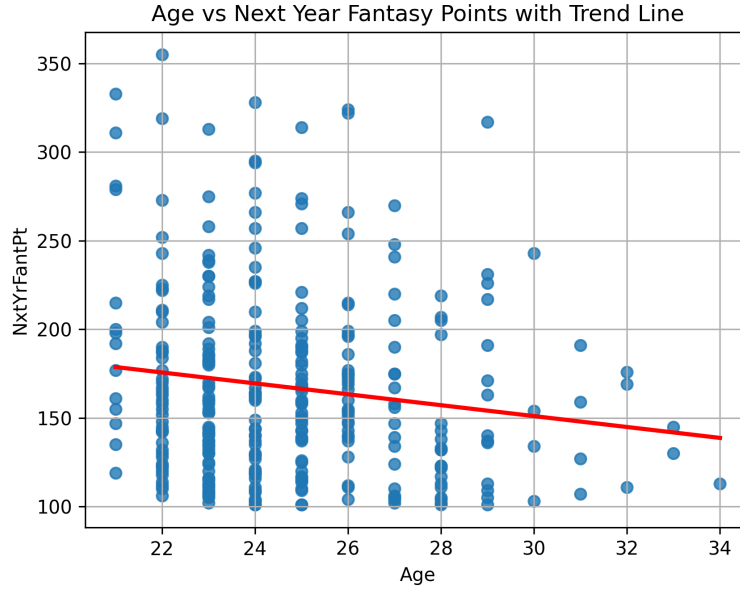


Figure 4: Age vs Next Year Fantasy Points

Rather than strict fantasy points, another idea was to look at how the models were able to predict the rankings of where they would finish. Looking at the linear model for simplicity, I ordered the players by their total fantasy points for the true next years and compared them to the predicted next years points. I then calculated an MSE of 520, or an RMSE of 22.8, which indicated this model was still not a good predictor of ranking.

From the above graphs, we can conclude that there is some minor trend between these variables and predicting the fantasy points gained next year. However, these statistics are not enough in providing the full picture on exactly how well the players will do in the next year.

5 Conclusion & Future Work

More data may be better when forming a picture in predicting the next years fantasy points. Players are not always in the same environment as they were in the previous year. They can change their team, which adds many more data points into the mix. For running backs, whether their offensive line is good or bad, or if the offensive coach schemes plays that highlight their talents may have a higher impact on their production. We also need to take into consideration injuries more so than just the amount of games that they play in a season. Different types of injuries can impact how a player performs in the following year, and how well they have recovered from it. However, how can we add something like emotions to the mix? If a team or coaches foster a positive environment, how much does that impact player performance aside from the stats? There are an extreme amount of data points that can be added, whether tangible or intangible.

In a test, I filtered out running backs with a 95% percentile discrepancy between prior and next year fantasy points. These players were ones that had a significant increase or decrease in their fantasy points over the year; whether due to injury or other circumstances. This helps in filtering out unpredictable injuries, however it could also filter out some breakout players. The models all improved slightly, seen below in Figure 4. This shows that there are more variables such as injury status that need to be taken into account to better the predictions.

RB Model Eval Metrics					
	Linear Regression	Random Forest	Boosting	K Nearest Neighbors	Neural Net
MSE	2111.272412	2173.841545	2162.596337	2727.495783	2135.563798
R2	0.641673	0.631054	0.632962	0.537087	0.637550
RMSE	45.948584	46.624474	46.503724	52.225432	46.212161

Figure 5: 95% RB Model Evaluation Metrics

After looking at how the models perform in predicting next years fantasy points with solely player statistics, we can see that they are not ideal. The margin for error is too great to be able to use them reliably in fantasy football drafts. Although some of the statistics do provide some insight, they are not enough to see the full picture. We cannot utilize solely this data in our analyses, but we can understand that it provides some insight and may be able to be combined with other data to give us a better model for prediction.

6 References

1. [https://en.wikipedia.org/wiki/Fantasy_football_\(gridiron\)](https://en.wikipedia.org/wiki/Fantasy_football_(gridiron))
2. https://www.espn.com/nfl/draft2025/story/_/id/44404897/nfl-draft-history-rules-traditions-mr-irrelevant-salaries-questions
3. <https://www.pro-football-reference.com/>